

Laporan Praktikum Praktikum 3
Pemrograman Mobile



Disusun oleh:
Muhammad Aditya (20240801234)

PROGRAM STUDI TEKNIK INFORMATIKA
FAKULTAS ILMU KOMPUTER
UNIVERSITAS ESA UNGGUL
2026

- Bagian A

```
import 'package:flutter/material.dart';
```

```
void main() => runApp(const MyApp());
```

```
class MyApp extends StatelessWidget {  
  const MyApp({super.key});
```

```
  @override
```

```
  Widget build(BuildContext context) {
```

```
    return MaterialApp(  
      title: 'Praktikum 3',
```

```
      theme: ThemeData(colorSchemeSeed: Colors.blue, useMaterial3: true),
```

```
      home: const InputPage(),
```

```
    );
```

```
  }
```

```
}
```

```
class InputPage extends StatefulWidget {
```

```
  const InputPage({super.key});
```

```
  @override
```

```
  State<InputPage> createState() => InputPageState();
```

```
}
```

```
class InputPageState extends State<InputPage> {
```

```
  final _controller = TextEditingController();
```

```
  String _salam = ";
```

```
  @override
```

```
  void dispose() {
```

```
    // Controller harus dilepas agar tidak terjadi kebocoran memori
```

```
    _controller.dispose();
```

```
    super.dispose();
```

```
  }
```

```
  @override
```

```
  Widget build(BuildContext context) {
```

```
    return Scaffold(  
      appBar: AppBar(title: const Text('Input Dasar')),
```

```
      body: Padding(  
        padding: const EdgeInsets.all(16),
```

```
        child: Column(  
          children: [
```

```
            TextField(  
              controller: _controller,
```

```
              decoration: const InputDecoration(  
                border: OutlineInputBorder(),
```

```
                hintText: 'Masukkan salam',
```

```
      labelText: 'Nama',
      border: OutlineInputBorder(),
    ),
  ),
  const SizedBox(height: 12),
  ElevatedButton(
    onPressed: () {
      setState(() => _salam = 'Halo, ${_controller.text}!');
    },
    child: const Text('Sapa'),
  ),
  const SizedBox(height: 12),
  Text(_salam, style: const TextStyle(fontSize: 20)),
],
),
),
);
}
```

22:45

100%

DEBUG

Tugas (0/1)

☐ pemmob



Sintaks	Penjelasan Fungsi
TextEditingController	Objek yang menyimpan dan membaca teks yang diketik pengguna di dalam TextField.
TextField	Widget kolom input teks; nilainya diambil lewat properti controller.
dispose()	Method yang dipanggil saat widget dihancurkan; controller wajib dilepas (dispose) di sini agar tidak terjadi kebocoran memori.
setState(() => _salam = ...)	Memperbarui variabel _salam sekaligus memberi tahu Flutter agar tampilan dibangun ulang.

- Bagian B

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';
```

```
// Definisi Model Data
```

```
class Tugas {
  String judul;
  bool selesai;
  Tugas(this.judul, {this.selesai = false});
}
```

```
// State Management menggunakan ChangeNotifier
```

```
class TugasModel extends ChangeNotifier {
  final List<Tugas> _items = [];
  List<Tugas> get items => List.unmodifiable(_items);
  int get jumlahSelesai => _items.where((t) => t.selesai).length;
```

```
  void tambah(String judul) {
    _items.add(Tugas(judul));
    notifyListeners();
  }
```

```
  void toggle(int index) {
    _items[index].selesai = !_items[index].selesai;
    notifyListeners();
  }
```

```
  void hapus(int index) {
    _items.removeAt(index);
    notifyListeners();
  }
}
```

```
void main() {
  runApp(
```

```

ChangeNotifierProvider(
  // PERBAIKAN UTAMA: Tambahkan parameter (context) pada fungsi create
  create: (context) => TugasModel(),
  child: const MyApp(),
),
);
}

```

```

class MyApp extends StatelessWidget {
  const MyApp({super.key});

```

```

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Daftar Tugas',
      theme: ThemeData(colorSchemeSeed: Colors.blue, useMaterial3: true),
      home: const TugasPage(),
    );
  }
}

```

// Halaman Daftar Tugas

```

class TugasPage extends StatelessWidget {
  const TugasPage({super.key});

```

```

  @override
  Widget build(BuildContext context) {
    final model = context.watch<TugasModel>();

```

```

    return Scaffold(
      appBar: AppBar(
        title: Text('Tugas (${model.jumlahSelesai}/${model.items.length}')),
      ),
      body: ListView.builder(
        itemCount: model.items.length,
        itemBuilder: (context, i) {
          final t = model.items[i];
          return ListTile(
            leading: Checkbox(
              value: t.selesai,
              onChanged: (_) => context.read<TugasModel>().toggle(i),
            ),
            title: Text(

```

```

        t.judul,
        style: TextStyle(
          decoration: t.selesai ? TextDecoration.lineThrough : null,
        ),
      ),
      trailing: IconButton(
        icon: const Icon(Icons.delete),
        onPressed: () => context.read<TugasModel>().hapus(i),
      ),
    );
  },
),
floatingActionButton: FloatingActionButton(
  onPressed: () {
    Navigator.push(
      context,
      MaterialPageRoute(builder: (_) => const TambahPage()),
    );
  },
  child: const Icon(Icons.add),
),
);
}
}

```

// Halaman Tambah Tugas

```

class TambahPage extends StatefulWidget {
  const TambahPage({super.key});

  @override
  State<TambahPage> createState() => _TambahPageState();
}

```

```

class _TambahPageState extends State<TambahPage> {
  final _controller = TextEditingController();

  @override
  void dispose() {
    _controller.dispose();
    super.dispose();
  }
}

```

```

void _simpan() {

```

```

    final judul = _controller.text.trim();
    if (judul.isEmpty) return;
    context.read<TugasModel>().tambah(judul);
    Navigator.pop(context);
}

```

```

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(title: const Text('Tambah Tugas')),
    body: Padding(
      padding: const EdgeInsets.all(16),
      child: Column(
        children: [
          TextField(
            controller: _controller,
            autofocus: true,
            decoration: const InputDecoration(
              labelText: 'Judul tugas',
              border: OutlineInputBorder(),
            ),
            onSubmitted: (_) => _simpan(),
          ),
          const SizedBox(height: 12),
          ElevatedButton(
            onPressed: _simpan,
            child: const Text('Simpan'),
          ),
        ],
      ),
    ),
  );
}

```

```

import 'package:flutter/material.dart';
import 'package:provider/provider.dart';

// Definisi Model Data
class Tugas {
  String judul;
  bool selesai;
  Tugas(this.judul, {this.selesai = false});
}

// State Management menggunakan ChangeNotifier
class TugasModel extends ChangeNotifier {
  final List<Tugas> _items = [];
  List<Tugas> get items => List.unmodifiable(_items);
  int get jumlahSelesai => _items.where((t) => t.selesai).length;

  void tambah(String judul) {
    _items.add(Tugas(judul));
    notifyListeners();
  }

  void toggle(int index) {
    _items[index].selesai = !_items[index].selesai;
    notifyListeners();
  }

  void hapus(int index) {
    _items.removeAt(index);
    notifyListeners();
  }
}

void main() {
  runApp(
    ChangeNotifierProvider(
      // PERBAIKAN UTAMA: Tambahkan parameter (context) pada fungsi create
      create: (context) => TugasModel(),
      child: const MyApp(),
    ),
  );
}

class MyApp extends StatelessWidget {

```

```

const MyApp({super.key});

@override
Widget build(BuildContext context) {
  return MaterialApp(
    title: 'Daftar Tugas',
    theme: ThemeData(colorSchemeSeed: Colors.blue, useMaterial3: true),
    home: const TugasPage(),
  );
}

// Halaman Daftar Tugas
class TugasPage extends StatelessWidget {
  const TugasPage({super.key});

  @override
  Widget build(BuildContext context) {
    final model = context.watch<TugasModel>();

    return Scaffold(
      appBar: AppBar(
        title: Text('Tugas (${model.jumlahSelesai}/${model.items.length}')),
      ),
      body: ListView.builder(
        itemCount: model.items.length,
        itemBuilder: (context, i) {
          final t = model.items[i];
          return ListTile(
            leading: Checkbox(
              value: t.selesai,
              onChanged: (_) => context.read<TugasModel>().toggle(i),
            ),
            title: Text(
              t.judul,
              style: TextStyle(
                decoration: t.selesai ? TextDecoration.lineThrough : null,
              ),
            ),
            trailing: IconButton(
              icon: const Icon(Icons.delete),
              onPressed: () => context.read<TugasModel>().hapus(i),
            ),
          );
        },
      ),
    );
  }
}

```

```

        );
      },
    ),
    floatingActionButton: FloatingActionButton(
      onPressed: () {
        Navigator.push(
          context,
          MaterialPageRoute(builder: (_) => const TambahPage()),
        );
      },
      child: const Icon(Icons.add),
    ),
  );
}
}

```

// Halaman Tambah Tugas

```

class TambahPage extends StatefulWidget {
  const TambahPage({super.key});

```

```

  @override
  State<TambahPage> createState() => _TambahPageState();
}

```

```

class _TambahPageState extends State<TambahPage> {
  final _controller = TextEditingController();

```

```

  @override
  void dispose() {
    _controller.dispose();
    super.dispose();
  }

```

```

  void _simpan() {
    final judul = _controller.text.trim();
    if (judul.isEmpty) return;
    context.read<TugasModel>().tambah(judul);
    Navigator.pop(context);
  }

```

```

  @override
  Widget build(BuildContext context) {
    return Scaffold(

```

```

appBar: AppBar(title: const Text('Tambah Tugas')),
body: Padding(
  padding: const EdgeInsets.all(16),
  child: Column(
    children: [
      TextField(
        controller: _controller,
        autofocus: true,
        decoration: const InputDecoration(
          labelText: 'Judul tugas',
          border: OutlineInputBorder(),
        ),
        onSubmitted: (_) => _simpan(),
      ),
      const SizedBox(height: 12),
      ElevatedButton(
        onPressed: _simpan,
        child: const Text('Simpan'),
      ),
    ],
  ),
),
);
}
}

```

- Tugas daftar belanja

```

import 'package:flutter/material.dart';
import 'package:provider/provider.dart';

```

```

// Model Data Belanja

```

```

class BarangBelanja {
  String nama;
  int jumlah;
  String kategori;
  bool dibeli;

```

```

  BarangBelanja({
    required this.nama,
    required this.jumlah,
    required this.kategori,
    this.dibeli = false,
  });
}

```

```

// State Management menggunakan ChangeNotifier
class BelanjaModel extends ChangeNotifier {
  final List<BarangBelanja> _items = [];

  List<BarangBelanja> get items => List.unmodifiable(_items);

  // Menghitung jumlah barang yang belum dibeli untuk ditampilkan di AppBar
  int get jumlahBelumDibeli => _items.where((b) => !b.dibeli).length;

  void tambahBarang(String nama, int jumlah, String kategori) {
    _items.add(BarangBelanja(nama: nama, jumlah: jumlah, kategori: kategori));
    notifyListeners();
  }

  void toggleDibeli(int index) {
    _items[index].dibeli = !_items[index].dibeli;
    notifyListeners();
  }

  void hapusBarang(int index) {
    _items.removeAt(index);
    notifyListeners();
  }
}

void main() {
  runApp(
    ChangeNotifierProvider(
      create: (context) => BelanjaModel(),
      child: const MyApp(),
    ),
  );
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Daftar Belanja',
      theme: ThemeData(colorSchemeSeed: Colors.green, useMaterial3: true),
      home: const DaftarBelanjaPage(),
    );
  }
}

```

```

// Halaman Daftar Belanja
class DaftarBelanjaPage extends StatelessWidget {
  const DaftarBelanjaPage({super.key});

  @override
  Widget build(BuildContext context) {
    final model = context.watch<BelanjaModel>();

    return Scaffold(
      appBar: AppBar(
        title: Text('Daftar Belanja (Belum: ${model.jumlahBelumDibeli})'),
      ),
      body: model.items.isEmpty
        ? const Center(
            child: Text(
              'Belum ada daftar belanja',
              style: TextStyle(fontSize: 18, color: Colors.grey),
            ),
          )
        : ListView.builder(
            itemCount: model.items.length,
            itemBuilder: (context, index) {
              final item = model.items[index];
              return Card(
                margin: const EdgeInsets.symmetric(horizontal: 12, vertical: 6),
                child: ListTile(
                  leading: Checkbox(
                    value: item.dibeli,
                    onChanged: (_) => context.read<BelanjaModel>().toggleDibeli(index),
                  ),
                  title: Text(
                    item.nama,
                    style: TextStyle(
                      decoration: item.dibeli ? TextDecoration.lineThrough : null,
                      fontWeight: FontWeight.bold,
                    ),
                  ),
                  subtitle: Text('Jumlah: ${item.jumlah} | Kategori: ${item.kategori}'),
                  trailing: IconButton(
                    icon: const Icon(Icons.delete, color: Colors.red),
                    onPressed: () => context.read<BelanjaModel>().hapusBarang(index),
                  ),
                ),
              ),
            );
          },
        ),
      ),
    );
  }
}

```

```

floatingActionButton: FloatingActionButton(
  onPressed: () {
    Navigator.push(
      context,
      MaterialPageRoute(builder: (_) => const TambahBelanjaPage()),
    );
  },
  child: const Icon(Icons.add),
),
);
}
}

// Halaman Form Tambah Belanja dengan Validasi
class TambahBelanjaPage extends StatefulWidget {
  const TambahBelanjaPage({super.key});

  @override
  State<TambahBelanjaPage> createState() => _TambahBelanjaPageState();
}

class _TambahBelanjaPageState extends State<TambahBelanjaPage> {
  final _formKey = GlobalKey<FormState>();
  final _namaController = TextEditingController();
  final _jumlahController = TextEditingController();
  String? _kategoriPilihan;

  final List<String> _kategoriList = ['Makanan', 'Minuman', 'Elektronik', 'Lainnya'];

  @override
  void dispose() {
    _namaController.dispose();
    _jumlahController.dispose();
    super.dispose();
  }

  void _simpan() {
    if (_formKey.currentState!.validate()) {
      final nama = _namaController.text.trim();
      final jumlah = int.parse(_jumlahController.text.trim());
      final kategori = _kategoriPilihan!;

      context.read<BelanjaModel>().tambahBarang(nama, jumlah, kategori);

      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(content: Text('Barang berhasil ditambahkan ke daftar belanja')),
      );
    }
  }
}

```

```

    Navigator.pop(context);
  }
}

```

```

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(title: const Text('Tambah Belanjaan')),
    body: Padding(
      padding: const EdgeInsets.all(16.0),
      child: Form(
        key: _formKey,
        child: ListView(
          children: [
            // Validasi Nama Barang
            TextFormField(
              controller: _namaController,
              decoration: const InputDecoration(
                labelText: 'Nama Barang',
                border: OutlineInputBorder(),
              ),
            ),
            validator: (v) {
              if (v == null || v.trim().isEmpty) {
                return 'Nama barang wajib diisi';
              }
              return null;
            },
          ),
          const SizedBox(height: 16),

          // Validasi Jumlah Barang (Angka > 0)
          TextFormField(
            controller: _jumlahController,
            keyboardType: TextInputType.number,
            decoration: const InputDecoration(
              labelText: 'Jumlah',
              border: OutlineInputBorder(),
            ),
          ),
          validator: (v) {
            if (v == null || v.trim().isEmpty) {
              return 'Jumlah wajib diisi';
            }
            final angka = int.tryParse(v);
            if (angka == null || angka <= 0) {
              return 'Jumlah harus berupa angka lebih dari 0';
            }
          }
        )
      )
    )
  );
}

```

```

        return null;
      },
    ),
    const SizedBox(height: 16),

    // Validasi Dropdown Kategori
    DropdownButtonFormField<String>(
      decoration: const InputDecoration(
        labelText: 'Kategori',
        border: OutlineInputBorder(),
      ),
      items: _kategoriList.map((kategori) {
        return DropdownMenuItem(
          value: kategori,
          child: Text(kategori),
        );
      }).toList(),
      onChanged: (v) {
        setState(() {
          _kategoriPilihan = v;
        });
      },
      validator: (v) => v == null ? 'Pilih kategori terlebih dahulu' : null,
    ),
    const SizedBox(height: 24),

    ElevatedButton(
      onPressed: _simpan,
      child: const Text('Simpan Barang'),
    ),
  ],
),
),
),
);
}
}

```

22:58

100%

Canva

← Tambah Belanjaan

Nama Barang

Indomie

Jumlah

2

Kategori

Makanan

Simpan Barang



22:58

100%

DEKUS

Daftar Belanja (Belum: 1)



Indomie

Jumlah: 2 | Kategori: Makanan

